

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE



DEEP LEARNING MINIPROJECTS

TITLE : AUTOMATED DEFECT DETECTION
IN MANUFACTURING

PROJECT MEMBER

BHARATHKUMAR J – 24ADR020

ERAMALINGAM S-24ADR045

ASHWIN J M-24ADL188



PROBLEM STATEMENT

- In modern manufacturing systems, maintaining consistent product quality is a critical challenge. Traditional inspection systems rely heavily on manual labor, which introduces several limitations.
- Human inspection is subjective and inconsistent.
- Difficulty in identifying micro-level defects such as cracks, scratches, or surface anomalies.
- Inspection speed is limited, affecting production efficiency.
- High operational cost due to continuous human involvement.

OBJECTIVE

- **To develop an automated inspection system:**
The system should be capable of analyzing product images captured from manufacturing lines and automatically.
- **To implement CNN-based feature extraction:**
Convolutional Neural Networks are used to automatically learn hierarchical features such as edges, textures, and patterns from images, eliminating the need for manual feature engineering.
- **To apply anomaly detection using normal data:**
Instead of relying on large labeled defect datasets, the model is trained only on normal (good) images, enabling it to learn the standard appearance of products and detect deviations as anomalies.

LITERATURE SURVEY

- **Traditional Methods (Gonzalez & Woods, 2008):** Used edge detection and thresholding but not effective for noisy and complex images
- **Machine Learning Methods (Vapnik, 1995; Breiman, 2001):** Models like SVM and Random Forest require manual feature extraction and depend on feature quality
- **Deep Learning Introduction (Krizhevsky et al., 2012):** CNN models like AlexNet improved accuracy by automatically learning features from images
- **Deep Learning Framework (Goodfellow et al., 2014):** Established deep learning concepts and widely used for image classification tasks

RESEARCH GAP

- **Needs large defect data:** Existing models require many defect images, which are difficult to collect .
- **Cannot detect new defects:** Models fail to identify unseen or unknown defect patterns
- **Sensitive to real-world changes:** Performance is affected by lighting, noise, and background variations
- **Accuracy issues:** Sometimes gives incorrect results such as false positives and false negatives
- **Not suitable for real-time use:** Difficult to implement in fast and dynamic industrial environments

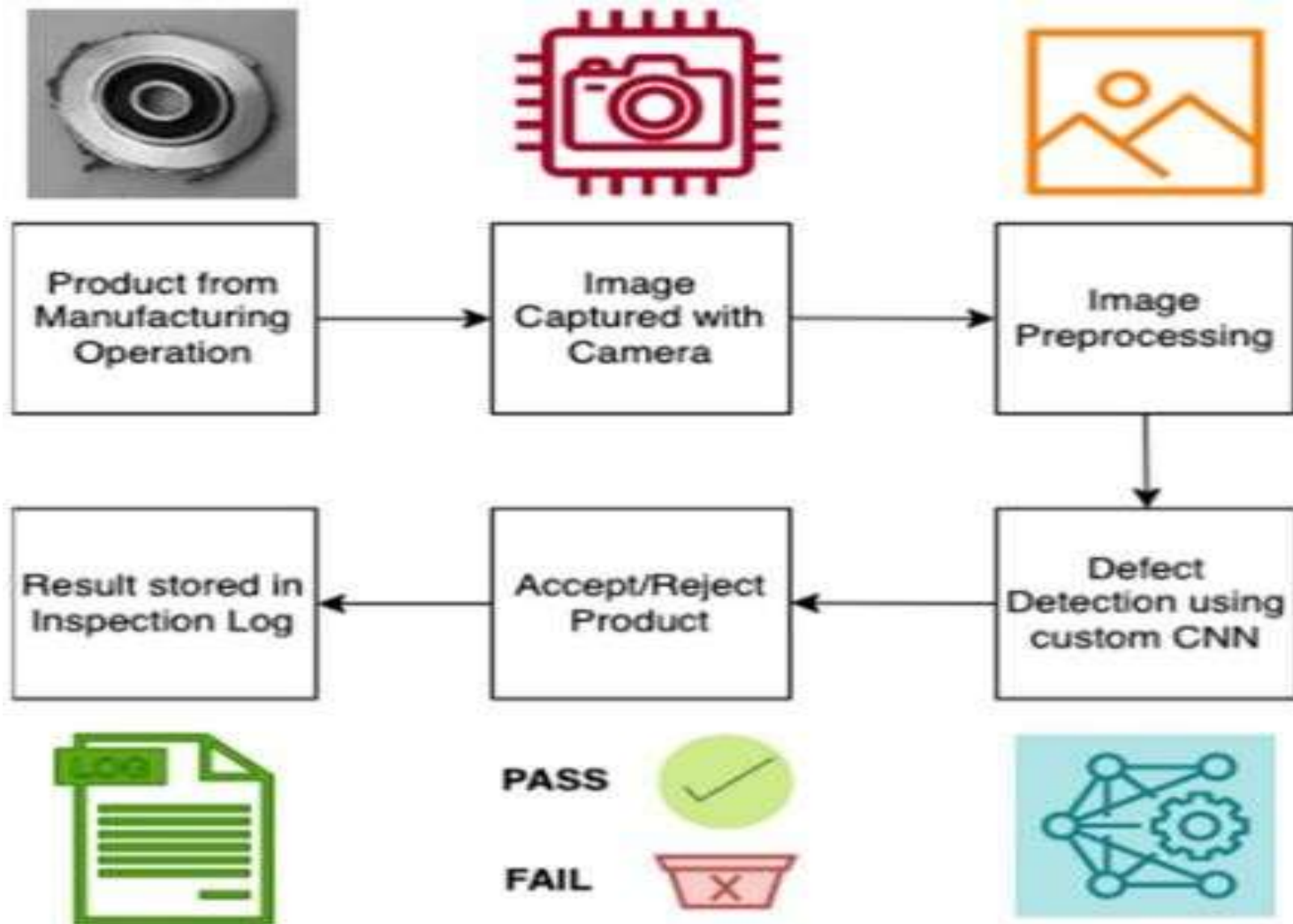
OBJECTIVE

- To develop an automated defect detection system for manufacturing industries that reduces manual inspection effort and improves consistency in quality control processes.
- To utilize Convolutional Neural Networks (CNNs) for automatic feature extraction from images, enabling the model to learn complex patterns such as textures and defects without manual intervention.
- To implement an anomaly detection approach by training the model only on normal (good) images, allowing the system to identify unseen defects by detecting deviations from learned patterns.
- To detect and classify defects based on reconstruction error, where a higher difference between the original and reconstructed image indicates the presence of defects.

PROPOSED WORK

- **To use the MVTec AD dataset for training and testing:** It provides real industrial images with normal and defective samples for model development
- **To preprocess the images for consistency:** All images are resized, normalized, and enhanced to improve model accuracy
- **To build a CNN-based autoencoder model:** The model learns important features and patterns from input images
- **To train the model using only normal (good) images:** This helps the system understand the standard appearance of products
- **To identify defects using reconstruction error:** The difference between original and reconstructed images indicates anomalies

WORK FLOW



SOFTWARE / HARDWARE REQUIREMENTS

Hardware Specifications:

- Processor: Intel Core i5 / Ryzen 5 or above.
- RAM: Minimum 8 GB (16 GB recommended).
- Storage: 256 GB SSD or above.
- Display: Standard monitor or laptop screen.

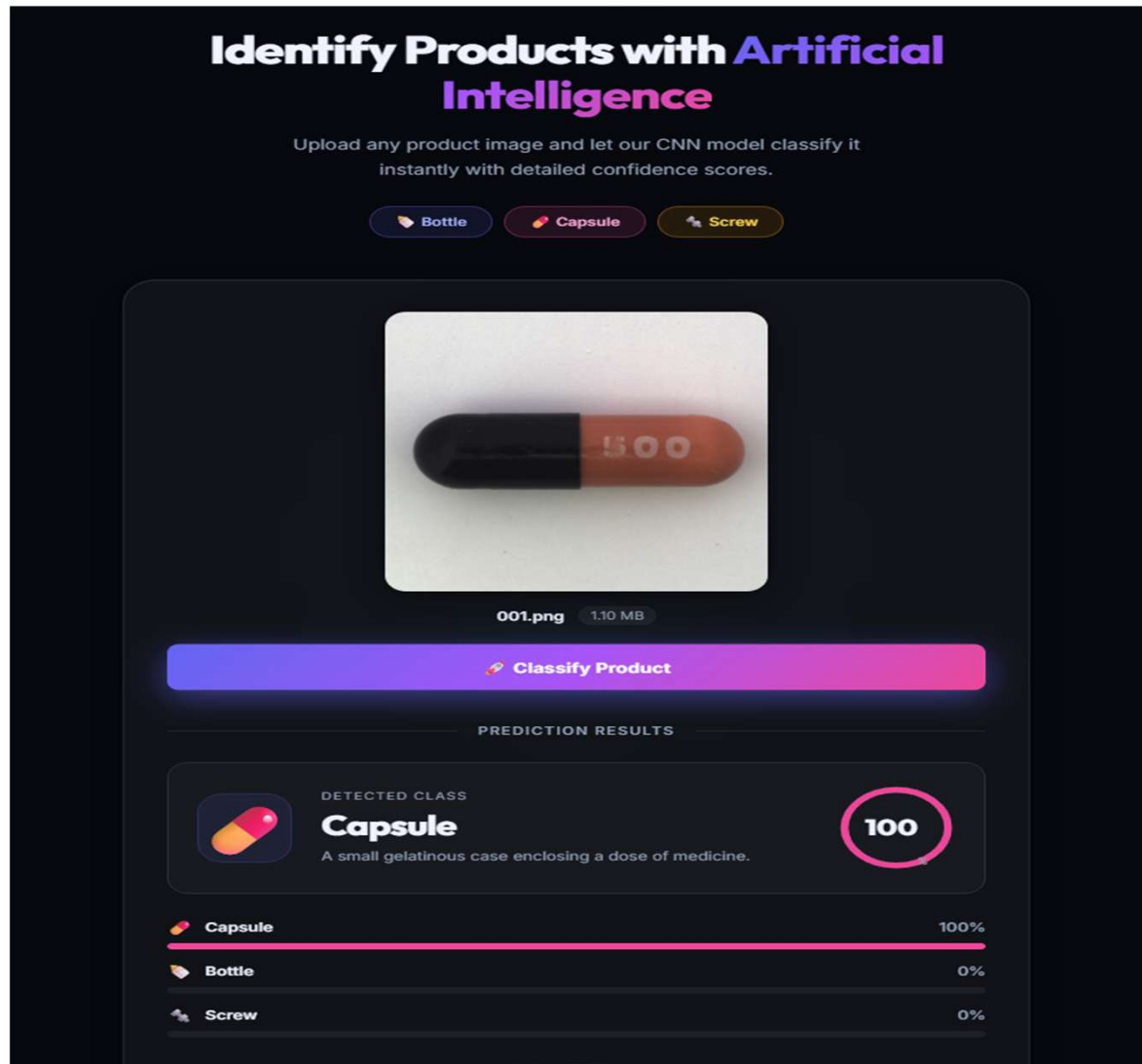
Software Stack:

- Programming Language: Python.
- Machine Learning Libraries: Scikit-learn, NumPy, Pandas.
- Web Framework: Flask.
- Model Saving: Pickle / Joblib.
- Development Tools: VS Code / Colab Notebook.
- Operating System: Windows / Linux.

MODULES WITH DESCRIPTION

- **Module 1 — Data Collection & Preprocessing:** Load MVTec dataset and prepare images using resizing, normalization, and data augmentation
- **Module 2 — Exploratory Data Analysis:** Analyze image data to understand normal and defect patterns and dataset characteristics
- **Module 3 — Model Training:** Train CNN-based autoencoder model using only normal images to learn standard features
- **Module 4 — Model Evaluation & Validation:** Evaluate model performance using reconstruction error and threshold on test data
- **Module 5 — Prediction System & Deployment:** Integrate trained model with web application and display output as defect or no defect

Project Screenshot



REFERENCES

- **Bergmann, P., Fauser, M., Sattlegger, D., & Steger, C. (2019):** *MVTec AD — A Comprehensive Real-World Dataset for Unsupervised Anomaly Detection*, IEEE Conference
- **Krizhevsky, A., Sutskever, I., & Hinton, G. (2012):** *ImageNet Classification with Deep Convolutional Neural Networks*, NIPS
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- **Ruff, L., et al. (2018):** *Deep One-Class Classification*, ICML
- **Schlegl, T., et al. (2017):** *Unsupervised Anomaly Detection with GANs*, IPMI
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- **Keras Documentation:** <https://keras.io>

THANK YOU

